Beyond Form Generators: Do Generative User Interfaces Have a Ceiling?

JO VERMEULEN, Autodesk Research, Canada

Generative User Interfaces (GenUI) are increasingly positioned as a paradigm shift toward outcome-oriented, AI-mediated interaction, yet many current examples demonstrate form-based, low-complexity tasks. In this position paper, I draw a parallel to earlier model-based automatic UI generation, which similarly promised abstraction and scalability but ultimately encountered a low ceiling that limited adoption. Using professional creative software like Maya, Premiere Pro, or Fusion 360 as an aspirational GenUI goal, I ask what it would mean for GenUI to operate at this level of complexity. What technical approaches would be required to achieve this? How might dynamically generated interfaces coexist with the human need for consistency, spatial memory, and long-term expertise development? And is scaling GenUI to match the complexity of these systems even the right trajectory, or will professional tools instead evolve toward agentic interaction layered on relatively constrained interfaces? By situating GenUI within HCI's historical debates on automation, adaptation, and direct manipulation versus agents, this position paper offers provocations for discussing the limits and future applications of GenUI in professional creative software.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**.

Additional Key Words and Phrases: Generative UI, creative software, design tools

1 Introduction

In this position paper, I propose a provocation for discussion at the workshop: **Do Generative User Interfaces (GenUI) have a ceiling?** Could GenUI scale to professional feature-rich creative design tools such as Premiere Pro, Maya, or Fusion 360? Do they need to? Are there technical or human limits? In what follows, I expand on these questions.

2 A Parallel with Model-Based Automatic User Interface Generation

Reflecting on the current momentum around Generative User Interfaces (GenUI) both in research [1] and emerging industry standards [2, 8, 9], I am reminded of some of the high hopes and promises of early work on model-based automatic generation of user interfaces [14]. This research area involved specifying user interface constraints and requirements using higher-level models such as task, domain, or dialog models and then automatically generating the final UI code from these models. With the availability of new mobile devices at the time that had varying operating systems and UI toolkits, there were important benefits to specifying the UI only once and generating it for each target platform [3]. This translation happened either with rule-based approaches or later on with ML-based techniques such as optimization [6]. This is conceptually similar to what Moran and Gibbons describe as the GenUI paradigm of *outcome-oriented design*, in which designers focus on user goals and define constraints (for AI to operate within), rather than design discrete interface elements [11].

3 The Form-Based Ceiling

While some studies showed that these early automatically generated interfaces can be superior in terms of usability or learnability [7, 13], many of these approaches placed severe limitations on the kinds of interfaces they could produce (what Myers et al. call a *low ceiling* [12]), in addition to other challenges such as unpredictability of the interfaces (a problem that we still deal with today with GenUI) and the difficulty of allowing designer input. These approaches excelled at generating interfaces with buttons, sliders, input fields, checkboxes or menus, but failed to scale to the complex and deeply specialized tools required for creative work. Eventually, these automatic generation approaches failed to

Author's Contact Information: Jo Vermeulen, Autodesk Research, Toronto, Canada, jo.vermeulen@autodesk.com.

gain widespread acceptance and traction in UX practice and were mostly superseded by designer-led workflows [13]. Looking at current GenUI standards and demos, we may be witnessing a similar trend, where most of the capabilities focus on relatively simple form-based interfaces and transactional tasks like booking a flight or checking a bank balance, or UIs that follow common formulaic patterns (like mobile apps). This raises the question of whether this reflects a recurrence of the same structural ceiling, albeit under a new technological paradigm.

If we consider highly sophisticated feature-rich professional applications such as Maya, 3ds Max, Fusion 360, DaVinci Resolve, Houdini, CATIA, or Premiere Pro, these often rely on precise direct manipulation, high information density, and workflows that take years to master. Could GenUI recreate an interface as sophisticated in terms of functionality and tooling as Premiere Pro or Maya? How can we scale GenUI beyond the *slider and button* paradigm to support the kind of complex direct manipulation required by professional tools? What new technical approaches will be needed?

4 Human Limits of GenUI

Assuming we can technically solve this and scale the complexity of GenUI up to reach the level of complex feature-rich professional software tools, will we then run into limitations in terms of usability and learnability? If a GenUI system generates a bespoke interface for every session or task, it may risk breaking the very spatial consistency that professional mastery requires. Will users get confused and frustrated by dynamically changing interfaces that are difficult to learn or support through peer learning and shared practice? Early work on adaptive and dynamic menus showed that inconsistency often caused significant issues [5, 10]. How can we reconcile the GenUI vision of radically different, tailor-made UIs with the human need for stability and muscle memory?

5 Direct Manipulation vs. Agents

Another perspective is that GenUI may not need to reach this level of interface complexity at all in an agentic AI future. This echoes prior debates on direct manipulation versus interface agents in the CHI community [4, 15]. Is the future of professional design software one where the user no longer manipulates pixels, vertices, waveforms, curves, or timelines directly, but instead describes the desired outcome to one or more agents? Will design software follow trends in software development with recent vibe coding tools that orchestrate swarms of specialized agents [16] that may generate transient dynamic UIs when needed?

6 Biography

Jo Vermeulen is a Senior Principal Research Scientist in the Human-Computer Interaction and Visualization group at Autodesk Research in Toronto. He holds a PhD in Computer Science from Hasselt University in Belgium, for which he received the 2015 IBM Innovation Award. He is currently interested in novel interfaces at the intersection of HCI and Generative AI, including Generative User Interfaces and agentic workflows.

References

[1] Yining Cao, Peiling Jiang, and Haijun Xia. 2025. Generative and Malleable User Interfaces with Generative and Evolving Task-Driven Data Model. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 686, 20 pages. doi:10.1145/3706598.3713285

[2] CopilotKit. 2026. Generative UI: Understanding Agent-Powered Interfaces. https://www.copilotkit.ai/generative-ui. Accessed: 2026-02-12.

[3] Jacob Eisenstein, Jean Vanderdonckt, and Angel Puerta. 2001. Applying model-based techniques to the development of UIs for mobile computers. In *Proceedings of the 6th International Conference on Intelligent User Interfaces* (Santa Fe, New Mexico, USA) *(IUI '01)*. Association for Computing Machinery, New York, NY, USA, 69–76. doi:10.1145/359784.360122

[4] Umer Farooq, Jonathan Grudin, Ben Shneiderman, Pattie Maes, and Xiangshi Ren. 2017. Human Computer Integration versus Powerful Tools. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) *(CHI EA '17)*. Association for Computing Machinery, New York, NY, USA, 1277–1282. doi:10.1145/3027063.3051137

[5] Leah Findlater and Joanna McGrenere. 2004. A comparison of static, adaptive, and adaptable menus. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) *(CHI '04)*. Association for Computing Machinery, New York, NY, USA, 89–96. doi:10.1145/985692.985704

[6] Krzysztof Gajos and Daniel S. Weld. 2004. SUPPLE: automatically generating user interfaces. In *Proceedings of the 9th International Conference on Intelligent User Interfaces* (Funchal, Madeira, Portugal) *(IUI '04)*. Association for Computing Machinery, New York, NY, USA, 93–100. doi:10.1145/964442.964461

[7] Krzysztof Z. Gajos, Jacob O. Wobbrock, and Daniel S. Weld. 2008. Improving the performance of motor-impaired users with automatically-generated, ability-based interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) *(CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1257–1266. doi:10.1145/1357054.1357250

[8] Google. 2026. A2UI. https://a2ui.org/. Accessed: 2026-04-09.

[9] MCP-UI Organization. 2026. MCP-UI: UI over MCP SDK. https://github.com/MCP-UI-Org/mcp-ui. Accessed 12 February 2026.

[10] Jeffrey Mitchell and Ben Shneiderman. 1989. Dynamic versus static menus: an exploratory comparison. *SIGCHI Bull.* 20, 4 (April 1989), 33–37. doi:10.1145/67243.67247

[11] Kate Moran and Sarah Gibbons. 2024. Generative UI and Outcome-Oriented Design. Nielsen Norman Group. https://www.nngroup.com/articles/generative-ui/ Accessed: February 13, 2026.

[12] Brad Myers, Scott E. Hudson, and Randy Pausch. 2000. Past, present, and future of user interface software tools. *ACM Trans. Comput.-Hum. Interact.* 7, 1 (March 2000), 3–28. doi:10.1145/344949.344959

[13] Jeffrey Nichols, Duen Horng Chau, and Brad A. Myers. 2007. Demonstrating the viability of automatically generated user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) *(CHI '07)*. Association for Computing Machinery, New York, NY, USA, 1283–1292. doi:10.1145/1240624.1240819

[14] Angel R. Puerta, Henrik Eriksson, John H. Gennari, and Mark A. Musen. 1994. Model-based automated generation of user interfaces. In *Proceedings of the Twelfth AAAI National Conference on Artificial Intelligence* (Seattle, Washington) *(AAAI'94)*. AAAI Press, 471–477.

[15] Ben Shneiderman and Pattie Maes. 1997. Direct manipulation vs. interface agents. *Interactions* 4, 6 (Nov. 1997), 42–61. doi:10.1145/267505.267514

[16] Steve Yegge. 2026. *Welcome to Gas Town*. Medium. https://steve-yegge.medium.com/welcome-to-gas-town-4f25ee16dd04 Accessed: 2026-02-12.